# 02_SSH Lab logs

Full process of setting public key auth onto SSH server

Github: github.com/Sceptic700

# Setup:

SSH server : Ubuntu 24.04.3 LTS (enp0s8: 192.168.0.1/24)
Client : Ubuntu 24.04.3 LTS (enp0s8: 192.168.0.2/24)

Networks (for both machines):
enp0s3 - NAT
enp0s8 - Internal network

# Task:

Install SSH , configure public key auth and provide public client key for the SSH server.

# Steps taken:

--- SSH server ---
0) "sudo apt install ssh"

--- Client ---

1) ssh-keygen -t ed25519

2) cp user_key /home/administrator/.ssh

Sending user public key to the SSH server
3) “scp user_key.pub administrator@192.168.0.1:/home/administrator”

--- SSH server ---

4) sudo cat user_key.pub > .ssh/authorized_keys

5)SSH requires some files to be protected by system:
“chmod 700 .ssh”
“chmod 600 .ssh/authorized_keys”
“sudo chown -R administrator:administrator .ssh”

--- Client ---

6) Same as step ‘5’:
“chmod 700 .ssh”
“chmod 600 .ssh/user_key”
“chmod 600 .ssh/authorized_keys”
“chown -R administrator:administrator .ssh”

7) ssh -i ~/.ssh/user_key -v administrator@192.168.0.1
- Connection succeeds

```
debug1: Next authentication method: publickey
debug1: Will attempt key: /home/administrator/.ssh/user_key ED25519 SHA256:mwqNBocgZMfCNSvNbT/4F/2vh4TAJcZkZ8KHPRRylTo explicit
debug1: Offering public key: /home/administrator/.ssh/user_key ED25519 SHA256:mwqNBocgZMfCNSvNbT/4F/2vh4TAJcZkZ8KHPRRylTo explicit
debug1: Server accepts key: /home/administrator/.ssh/user_key ED25519 SHA256:mwqNBocgZMfCNSvNbT/4F/2vh4TAJcZkZ8KHPRRylTo explicit
Authenticated to 192.168.0.1 ([192.168.0.1]:22) using "publickey".
debug1: channel 0: new session [client-session] (inactive timeout: 0)
debug1: Requesting no-more-sessions@openssh.com
debug1: Entering interactive session.
debug1: pledge: filesystem
debug1: client_input_global_request: rtype hostkeys-00@openssh.com want_reply 0
debug1: client_input_hostkeys: searching /home/administrator/.ssh/known_hosts for 192.168.0.1 / (none)
debug1: client_input_hostkeys: searching /home/administrator/.ssh/known_hosts2 for 192.168.0.1 / (none)
debug1: client_input_hostkeys: hostkeys file /home/administrator/.ssh/known_hosts2 does not exist
debug1: client_input_hostkeys: no new or deprecated keys from server
debug1: Remote: /home/administrator/.ssh/authorized_keys:1: key options: agent-forwarding port-forwarding pty user-rc x11-forwarding
debug1: Remote: /home/administrator/.ssh/authorized_keys:1: key options: agent-forwarding port-forwarding pty user-rc x11-forwarding
debug1: Sending environment.
debug1: channel 0: setting env LANG = "en_US.UTF-8"
debug1: pledge: fork
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-85-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Dec 29 11:13:49 PM UTC 2025

  System load:             0.0
  Usage of /:              14.1% of 24.44GB
  Memory usage:            6%
  Swap usage:              0%
  Processes:               113
  Users logged in:         1
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe55:acab


Expanded Security Maintenance for Applications is not enabled.

85 updates can be applied immediately.
33 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Mon Dec 29 22:51:45 2025 from 192.168.0.2
administrator@ubungu:~$ _
```

8) For more automatic connect , so we don't write private key location each time,
we can write client side config file.
touch .ssh/config

```
  GNU nano 7.2
Host 192.168.0.1
    User administrator
    IdentityFile ~/.ssh/user_key
```

9) Now we can connect just by writing:
ssh 192.168.0.1

```
administrator@ubungu:~$ ssh 192.168.0.1
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-85-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Dec 31 12:36:33 AM UTC 2025

  System load:             0.0
  Usage of /:              14.2% of 24.44GB
  Memory usage:            6%
  Swap usage:              0%
  Processes:               110
  Users logged in:         1
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe55:acab


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Enable ESM Apps to receive additional future security updates.
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Last login: Mon Dec 29 23:13:50 2025 from 192.168.0.2
administrator@ubungu:~$
```

--- SSH server ---

10) For better security we can now turn off password auth.
"sudo nano /etc/ssh/sshd_config"
PasswordAuthentication no

```
#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication no_
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
```

--- Client ---

11) ssh -v 192.168.0.1
- Now the only way to log in is public key auth

```
debug1: Authentications that can continue: publickey
debug1: Next authentication method: publickey
debug1: Will attempt key: /home/administrator/.ssh/user_key ED255
debug1: Offering public key: /home/administrator/.ssh/user_key E
debug1: Server accepts key: /home/administrator/.ssh/user_key ED2
Authenticated to 192.168.0.1 ([192.168.0.1]:22) using "publickey"
debug1: channel 0: new session [client-session] (inactive timeout
debug1: Requesting no-more-sessions@openssh.com
debug1: Entering interactive session.
debug1: pledge: filesystem
debug1: client_input_global_request: rtype hostkeys-00@openssh.co
debug1: client_input_hostkeys: searching /home/administrator/.ssh
debug1: client_input_hostkeys: searching /home/administrator/.ssh
debug1: client_input_hostkeys: hostkeys file /home/administrator/
debug1: client_input_hostkeys: no new or deprecated keys from ser
debug1: Remote: /home/administrator/.ssh/authorized_keys:1: key o
debug1: Remote: /home/administrator/.ssh/authorized_keys:1: key o
debug1: Sending environment.
debug1: channel 0: setting env LANG = "en_US.UTF-8"
debug1: pledge: fork
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-85-generic x86_64
```